Run report: Orchestral Input Pipeline: Structured Prompt Conditioning for Full-Ensemble AI Music Generation

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Conducted independently; not an official Brunel University London project.

Run: 20261030-220005_full_7a8b2b82 (full mode)

Started 2026-10-30T22:00:05+00:00, finished 2026-10-30T22:04:24+00:00

Config SHA-256: 7a8b2b82226ba1fe0b6ebe6120b84108140d8ad4df0a166e24f1cc3137a80a1a

Requests SHA-256: 78b651e6688c6520c115aa3448807a3faaf9803ee6078ecc7780dc60f1f6266c

Pieces evaluated: 16 (failed: 0)

Human participants: none (fully computational study)

Every file in this folder is listed with its SHA-256 hash in SHA256SUMS.

Results in plain words

midi: prompt_adherence could not be tested (2 pairs; the test needs at least 6).

midi: section_clarity could not be tested (2 pairs; the test needs at least 6).

midi: dynamics_accuracy could not be tested (2 pairs; the test needs at least 6).

midi: structure_f could not be tested (2 pairs; the test needs at least 6).

musicgen: prompt_adherence could not be tested (2 pairs; the test needs at least 6).

musicgen: section_clarity could not be tested (2 pairs; the test needs at least 6).

musicgen: dynamics_accuracy could not be tested (2 pairs; the test needs at least 6).

musicgen: structure_f could not be tested (2 pairs; the test needs at least 6).

stable_audio: prompt_adherence could not be tested (2 pairs; the test needs at least 6).

stable_audio: section_clarity could not be tested (2 pairs; the test needs at least 6).

stable_audio: dynamics_accuracy could not be tested (2 pairs; the test needs at least 6).

stable_audio: structure_f could not be tested (2 pairs; the test needs at least 6).

Mean scores by generator and condition

generator	condition	prompt_adherence	section_clarity	dynamics_accuracy	structure_f	structure_chance	judge_overall	dynamic_range_db	range_violations
midi	pipeline	0.008	0.518	0.952	0.733	0.761	7.000	20.976	0.000
midi	plain	-0.005	0.339	-0.400	0.833	0.761	7.000	2.164	38.500
musicgen	expert	0.009	0.277	0.008	0.733	0.761	7.000	14.279	-
musicgen	pipeline	0.010	0.268	0.408	0.633	0.761	7.000	5.769	-
musicgen	plain	0.003	0.714	0.358	0.833	0.761	7.000	12.450	-
stable_audio	expert	0.005	0.312	-0.655	0.733	0.761	7.000	39.289	-
stable_audio	pipeline	0.008	0.375	-0.406	0.900	0.761	7.000	32.505	-
stable_audio	plain	0.007	0.219	-0.442	0.900	0.761	7.000	38.418	-

structure_chance is the boundary score expected from random boundaries; structure_f should be read against it.

Confirmatory tests (preregistered)

family	generator	comparison	metric	n_pairs	median_a	median_b	median_diff	ci95_low	ci95_high	W	p	p_holm	significant	rank_biserial
confirmatory	midi	pipeline vs plain	prompt_adherence	2	0.008	-0.005	0.014	0.005	0.023	-	-	-	None	1.000
confirmatory	midi	pipeline vs plain	section_clarity	2	0.518	0.339	0.179	-0.143	0.500	-	-	-	None	0.333

family	generator	comparison	metric	n_pairs	median_a	median_b	median_diff	ci95_low	ci95_high	W	p	p_holm	significant	rank_biserial
confirmatory	midi	pipeline vs plain	dynamics_accuracy	2	0.952	-0.400	1.352	1.176	1.527	-	-	-	None	1.000
confirmatory	midi	pipeline vs plain	structure_f	2	0.733	0.833	-0.100	-0.200	0.000	-	-	-	None	-1.000
confirmatory	musicgen	pipeline vs plain	prompt_adherence	2	0.010	0.003	0.006	0.00046	0.012	-	-	-	None	1.000
confirmatory	musicgen	pipeline vs plain	section_clarity	2	0.268	0.714	-0.446	-0.750	-0.143	-	-	-	None	-1.000
confirmatory	musicgen	pipeline vs plain	dynamics_accuracy	2	0.408	0.358	0.050	-0.733	0.833	-	-	-	None	0.333
confirmatory	musicgen	pipeline vs plain	structure_f	2	0.633	0.833	-0.200	-0.400	0.000	-	-	-	None	-1.000
confirmatory	stable_audio	pipeline vs plain	prompt_adherence	2	0.008	0.007	0.002	-0.000553	0.004	-	-	-	None	0.333
confirmatory	stable_audio	pipeline vs plain	section_clarity	2	0.375	0.219	0.156	0.000	0.312	-	-	-	None	1.000
confirmatory	stable_audio	pipeline vs plain	dynamics_accuracy	2	-0.406	-0.442	0.036	-0.145	0.218	-	-	-	None	0.333
confirmatory	stable_audio	pipeline vs plain	structure_f	2	0.900	0.900	0.000	0.000	0.000	-	-	-	None	0.000

Secondary tests

family	generator	comparison	metric	n_pairs	median_a	median_b	median_diff	ci95_low	ci95_high	W	p	p_holm	significant	rank_biserial
secondary	musicgen	pipeline vs expert	prompt_adherence	2	0.010	0.009	0.000472	-0.000129	0.001	-	-	-	None	0.333

family	generator	comparison	metric	n_pairs	median_a	median_b	median_diff	ci95_low	ci95_high	W	p	p_holm	significant	rank_biserial
secondary	musicgen	pipeline vs expert	section_clarity	2	0.268	0.277	-0.009	-0.143	0.125	-	-	-	None	-0.333
secondary	musicgen	pipeline vs expert	dynamics_accuracy	2	0.408	0.008	0.400	-0.033	0.833	-	-	-	None	0.333
secondary	musicgen	pipeline vs expert	structure_f	2	0.633	0.733	-0.100	-0.200	0.000	-	-	-	None	-1.000
secondary	musicgen	pipeline vs expert	judge_overall	2	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	musicgen	pipeline vs expert	cultural_similarity	1	0.013	0.012	0.000993	-	-	-	-	-	None	1.000
secondary	musicgen	pipeline vs expert	dynamic_range_db	2	5.769	14.279	-8.510	-8.666	-8.354	-	-	-	None	-1.000
secondary	musicgen	expert vs plain	prompt_adherence	2	0.009	0.003	0.006	-0.000613	0.012	-	-	-	None	0.333
secondary	musicgen	expert vs plain	section_clarity	2	0.277	0.714	-0.438	-0.875	0.000	-	-	-	None	-1.000
secondary	musicgen	expert vs plain	dynamics_accuracy	2	0.008	0.358	-0.350	-1.567	0.867	-	-	-	None	-0.333
secondary	musicgen	expert vs plain	structure_f	2	0.733	0.833	-0.100	-0.200	0.000	-	-	-	None	-1.000
secondary	musicgen	expert vs plain	judge_overall	2	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	musicgen	expert vs plain	cultural_similarity	1	0.012	0.013	-0.00058	-	-	-	-	-	None	-1.000
secondary	musicgen	expert vs plain	dynamic_range_db	2	14.279	12.450	1.828	-6.967	10.624	-	-	-	None	0.333

family	generator	comparison	metric	n_pairs	median_a	median_b	median_diff	ci95_low	ci95_high	W	p	p_holm	significant	rank_biserial
secondary	stable_audio	pipeline vs expert	prompt_adherence	2	0.008	0.005	0.003	-7.85e-05	0.007	-	-	-	None	0.333
secondary	stable_audio	pipeline vs expert	section_clarity	2	0.375	0.312	0.062	0.000	0.125	-	-	-	None	1.000
secondary	stable_audio	pipeline vs expert	dynamics_accuracy	2	-0.406	-0.655	0.248	0.158	0.339	-	-	-	None	1.000
secondary	stable_audio	pipeline vs expert	structure_f	2	0.900	0.733	0.167	0.000	0.333	-	-	-	None	1.000
secondary	stable_audio	pipeline vs expert	judge_overall	2	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	stable_audio	pipeline vs expert	cultural_similarity	1	0.010	0.010	-7.26e-05	-	-	-	-	-	None	-1.000
secondary	stable_audio	pipeline vs expert	dynamic_range_db	2	32.505	39.289	-6.785	-10.059	-3.510	-	-	-	None	-1.000
secondary	stable_audio	expert vs plain	prompt_adherence	2	0.005	0.007	-0.002	-0.003	-0.000474	-	-	-	None	-1.000
secondary	stable_audio	expert vs plain	section_clarity	2	0.312	0.219	0.094	0.000	0.188	-	-	-	None	1.000
secondary	stable_audio	expert vs plain	dynamics_accuracy	2	-0.655	-0.442	-0.212	-0.303	-0.121	-	-	-	None	-1.000
secondary	stable_audio	expert vs plain	structure_f	2	0.733	0.900	-0.167	-0.333	0.000	-	-	-	None	-1.000
secondary	stable_audio	expert vs plain	judge_overall	2	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	stable_audio	expert vs plain	cultural_similarity	1	0.010	0.011	-0.000476	-	-	-	-	-	None	-1.000

family	generator	comparison	metric	n_pairs	median_a	median_b	median_diff	ci95_low	ci95_high	W	p	p_holm	significant	rank_biserial
secondary	stable_audio	expert vs plain	dynamic_range_db	2	39.289	38.418	0.872	-0.797	2.540	-	-	-	None	0.333
secondary	midi	pipeline vs plain	judge_overall	2	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	midi	pipeline vs plain	cultural_similarity	1	0.013	0.008	0.005	-	-	-	-	-	None	1.000
secondary	midi	pipeline vs plain	dynamic_range_db	2	20.976	2.164	18.812	18.300	19.324	-	-	-	None	1.000
secondary	midi	pipeline vs plain	judge_overall	2	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	musicgen	pipeline vs plain	judge_overall	2	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	musicgen	pipeline vs plain	cultural_similarity	1	0.013	0.013	0.000413	-	-	-	-	-	None	1.000
secondary	musicgen	pipeline vs plain	dynamic_range_db	2	5.769	12.450	-6.681	-15.321	1.958	-	-	-	None	-0.333
secondary	musicgen	pipeline vs plain	judge_overall	2	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	stable_audio	pipeline vs plain	judge_overall	2	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000
secondary	stable_audio	pipeline vs plain	cultural_similarity	1	0.010	0.011	-0.000548	-	-	-	-	-	None	-1.000
secondary	stable_audio	pipeline vs plain	dynamic_range_db	2	32.505	38.418	-5.913	-7.519	-4.306	-	-	-	None	-1.000
secondary	stable_audio	pipeline vs plain	judge_overall	2	7.000	7.000	0.000	0.000	0.000	-	-	-	None	0.000

Agreement between measures (Spearman)

measure_a	measure_b	n	spearman_rho	p
prompt_adherence	section_clarity	16	-0.384	0.142
prompt_adherence	dynamics_accuracy	16	0.065	0.812
prompt_adherence	structure_f	16	-0.492	0.053
section_clarity	dynamics_accuracy	16	0.221	0.410
section_clarity	structure_f	16	0.034	0.902
dynamics_accuracy	structure_f	16	-0.091	0.739

Intent-parser accuracy on the benchmark

Mean family F1 1.000, form correct 2/2, dynamic shape correct 2/2 (optimistic: rules were written with these requests).

Held-out requests: mean family F1 0.703, form correct 12/15, shape correct 3/15.

Frechet Audio Distance (CLAP embeddings, lower is closer to real recordings)

No rows.

Limitations

All measures are automatic proxies for musical quality; no human listeners were involved.

The AI judge is itself a model with its own biases; it is reported as a secondary measure only.

CLAP similarities depend on the CLAP model's training data, which under-represents some styles (for example Indian classical music).

Pieces are short (default 30 s), so large-scale form is tested only in compressed form.

The intent parser was developed alongside the main benchmark; its accuracy there is optimistic. Parser accuracy on the held-out requests is the fair estimate.

Closed services (Suno) cannot be re-run exactly and are excluded from confirmatory tests.

Reproduce and verify

python run_all.py verify results/20261030-220005_full_7a8b2b82